EMC ASSESSMENT (UKCA)

Product: ST2VGA Adapter

Model/Revision: ST2VGA Adapter Rev.3 / REV3.4.0

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1. PRODUCT OVERVIEW

The product is a low-voltage electronic device intended for use in retro computing systems. It is designed as an external or semi-external component (e.g. standalone adapter, dongle, or external interface module) and is powered either by the host system or by an external low-voltage DC supply.

Typical operating voltage is below 50 V DC (e.g. 3.3V / 5V / 12V).

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2. APPLICABLE REGULATIONS

This assessment is made in relation to:

- Electromagnetic Compatibility Regulations 2016 (UK SI 2016 No. 1091)

The product does not fall under:

- Electrical Equipment (Safety) Regulations 2016
  (as it operates below 50 V DC)

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3. DESIGN CHARACTERISTICS RELEVANT TO EMC

- The device contains standard digital and/or analog electronic components (e.g. CMOS/TTL logic, microcontrollers, analog amplifiers, memory devices).
- No radio transmitter or intentional RF radiator is present.
- The device does not include wireless communication functionality.
- The design does not incorporate high-frequency switching power supplies.
- Signal frequencies are limited to those typical of legacy computing platforms and their peripheral interfaces.
- External cables (signal and power) are of limited length, are used in typical retro computing environments, and may act as potential emission paths.
- The device processes and amplifies analog RGB video signals and drives an external VGA interface.
- Signal frequencies correspond to legacy video timings (e.g. ~15 kHz and ~31 kHz horizontal sync rates), which are relatively low compared to modern high-speed digital interfaces.

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4. EMISSION ASSESSMENT

Due to the design characteristics described above:

- The product is not expected to generate significant electromagnetic emissions.
- Any emissions are comparable to those of similar small retro-computing accessories.
- The absence of high-frequency switching regulators and RF transmitters significantly reduces emission risk.

- The analog nature of the video signals reduces the likelihood of significant radiated emissions compared to high-speed digital interfaces.
- External cabling is limited in length and used in typical retro computing environments.
- Video timing frequencies are well below modern digital bus speeds, further reducing radiated emission risk.

Conclusion:
The product presents a low risk of causing electromagnetic interference when used as intended.

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5. IMMUNITY ASSESSMENT

- The product is designed to operate in typical residential, commercial, and light-industrial electromagnetic environments.
- Immunity performance depends on the overall installation and on the quality of the external DC supply and connected cabling.
- The product does not include specific shielding or filtering beyond standard design practices.

Conclusion:
The product is expected to operate correctly in typical residential, commercial, and light-industrial environments when used with compatible host equipment and a compliant external DC supply.

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6. INSTALLATION AND USE CONDITIONS

- The product is intended to be connected externally between compatible host equipment and its peripherals.
- It operates as a self-contained adapter or dongle; it is not part of the host's internal circuitry.
- Where the product is powered by an external DC supply (e.g. USB power adapter), that supply is outside the scope of this assessment and must itself carry the UKCA mark. Where the product is powered from the host system, no external supply falls within the scope of this assessment.
- Proper operation and EMC performance depend on correct cabling and use with compatible host equipment.

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7. OVERALL CONCLUSION

Based on the design characteristics and intended use, the product is considered to comply with the essential requirements of the Electromagnetic Compatibility Regulations 2016, provided that it is installed and used in accordance with its intended purpose.

This assessment is supported by the use of designated standards such as:

- BS EN IEC 61000-6-1:2019 (Immunity)

- BS EN IEC 61000-6-3:2021 (Emissions)

No additional EMC mitigation measures are required based on the design and intended use.

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8. RESPONSIBLE PERSON

Name: Diego Parrilla Santamaria
Position: CEO
Company: GOODDATA LABS SL

Date: 23 April 2026

Signature: